Exercise: Pneumatic Machine Control
Candy Packaging Machine

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1 Scenario

Imagine you are tasked with designing a pneumatic control system for an automated packaging machine. This machine is responsible for packaging small items (e.g., candies) into individual boxes. The machine has four pneumatic actuators:

1. Actuator 1 (**A**): Responsible for holding the suction cup gripper.
2. Suction cup 1 (B): Responsible for picking up items from a conveyor belt.
3. Actuator 2 (**C**): Responsible for opening and closing the box.
4. Actuator 3 (**D**): Responsible for rotating **A** to lift candies from the belt.
5. Actuator 4 (**E**): Responsible for sealing the box.

2 Objective

Your objective is to design a pneumatic control system, using GRAFCET that achieves the following sequence of operations:

2.1 Sequence

1. **D** rotate **A**, positioning it to pick up an candy.
2. **A** extends to prepare the gripper.

3. **B** engage to pick up an item.
4. **A** retracts.
5. **D** rotate **A**, positioning it to place an candy in the box.
6. **C** opens the box.
7. **A** extends, placing the item into the open box.
8. **B** disengage to realese the candy.
9. **A** retracts.
10. **C** closes the box.
11. **E** extends to seal the closed box.